National University of Computer & Emerging Sciences, Karachi
Fall-2023 Fast School of Computing
Mid-II Examination
9 November 2023, 08:30 AM – 09:30 AM

Course Code: MT1004	Course Name: Linear Algebra
Instructor(s) Name: Alishba Tariq, Fareeha Sultan, Dr. Khusro Mian, Mairaj Ahmed, Moheez ur Rahim, Shahid Ashraf, Usama Antuley,	
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Instructions:

- Attempt all questions. There are **03 Questions and 02 pages**.
- Solve the paper according to the **sequence given** in the question paper.
- Graphical Calculator is not allowed.
- Return the question paper with the answer copy.

Time: 60 minutes **Max Marks: 30 points**

Question 01: CLO-2 [5+5]

a) Answer the following

i) The set of all real-valued functions f defined everywhere on the real line and such that $f(2) = 0$ with the standard operations is a vector space. State yes or no. __________

ii) If A is a 3X4 matrix and its row space dimension is 2 than nullity of A is __________.

iii) How many vectors are in the standard basis for the vector space $P_5(all\ polynomials\ of\ degree \leq 5)$? __________

iv) If u is the linear combination of v_1 and v_2 then $\{u, v_1, v_2\}$ is linearly independent. State yes or no. __________

v) If eigen values of A are $\lambda = -1, -2, -1, 3, 2$ then algebraic multiplicities of all eigen values of A are ____________respectively.

b) Choose the correct answer(s)

i) Which of the following vectors in R^3 does not lie on the same line as the others?
(A) (1, −7, 4) (B) (−4, −28, 16) (C) (−3, 21, −12) (D) (2, −14, 8)

ii) Given the basis B ={(4, 0),(0, 1)} and the vector w =(2, 3), what is the coordinate vector $[w]_B$?
(A) $(\frac{1}{2}, 3)$ (B) (8,3) (C) (2,6) (D) (2,3)

iii) For a consistent system $AX = b$, b must lie in ____________.
(A) column space of A (B) Row space of A (C) Null space of A (D) none of these

iv) Which of the following is not an eigenvector of the matrix $\begin{bmatrix} 3 & 0 \\ 1 & 9 \end{bmatrix}$?
(A) $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ (B) $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$ (C) $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$ (D) $\begin{bmatrix} -6 \\ 1 \end{bmatrix}$

v) Sum of all eigen values of A is equal to __________.
(A) trace(A) (B) trace(A^T) (C) trace(A^{-1}) (D) All of these

Question 02: CLO-2 [6+4]

a) Consider the matrix A,

$$A = \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ -3 & 6 & -1 & 0 & -7 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

Find

i. the basis for row and column space of A.

ii. the basis for null space of A.

iii. *Verify dimension theorem.*

b) For which real values of λ do the following vectors form a linearly dependent set in R^3?

$v_1 = (\lambda, -\frac{1}{2}, -\frac{1}{2})$, $v_2 = (-\frac{1}{2}, \lambda, -\frac{1}{2})$, $v_3 = (-\frac{1}{2}, -\frac{1}{2}, \lambda)$

Question 03: CLO-2 [3+2+4+1]

Consider the matrix $A = \begin{bmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}$

a) Find the characteristic polynomial for the above matrix.

b) Find the eigen values for the above matrix.

c) Find the eigen vector(s) corresponding to all eigen values which are obtained in part(b).

d) Verify$\lambda_1 \lambda_2 \lambda_3 = \det(A)$